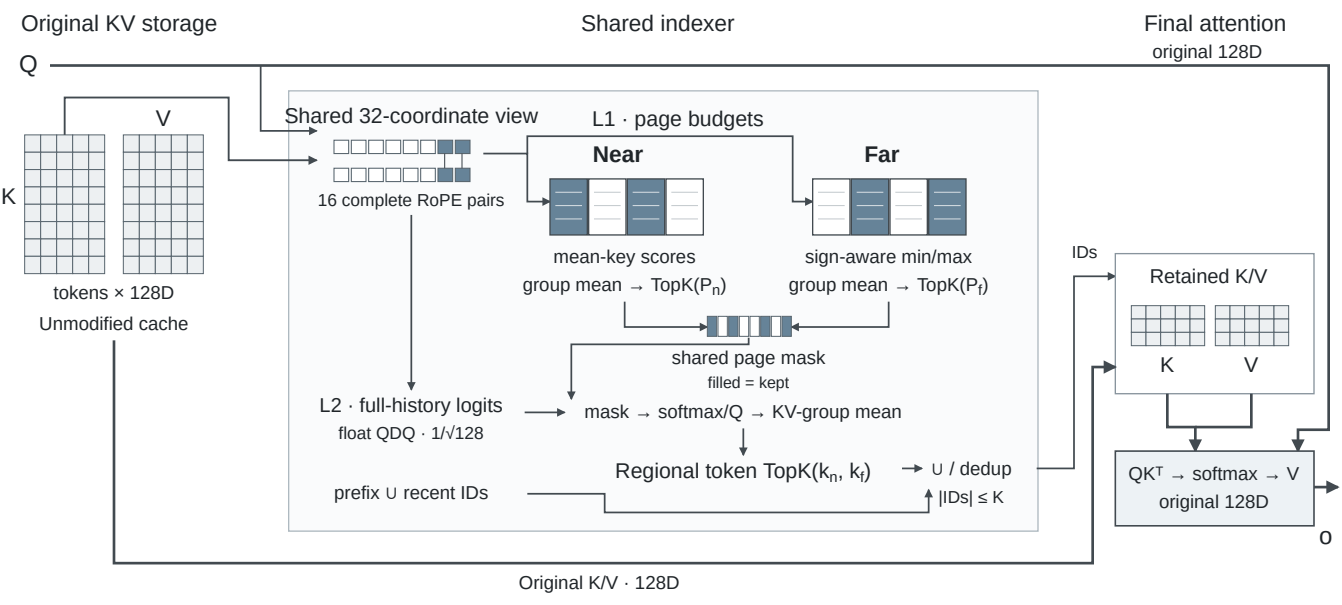
Original KV storage
Shared indexer
Final attention
original 128D
Q
V
K
tokens × 128D
Unmodified cache
Shared 32-coordinate view
16 complete RoPE pairs
L1 · page budgets
Near
Far
mean-key scores
group mean → TopK(P_n)
sign-aware min/max
group mean → TopK(P_f)
shared page mask
filled = kept
L2 · full-history logits
float QDQ · 1/√128
mask → softmax/Q → KV-group mean
Regional token TopK(k_n, k_f)
∪ / dedup
|IDs| ≤ K
prefix ∪ recent IDs
IDs
Retained K/V
K
V
QK^T → softmax → V
original 128D
o
Original K/V · 128D